

ZACHARY E ZEIDLER, Ph.D

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EDUCATION & TRAINING

- 2020 – present **University of California, Los Angeles, Los Angeles CA**
Postdoctoral Fellow, Physiology Department
Advisor: Laura DeNardo, Ph.D.
Prefrontal dynamics in learning and memory: I identified a cellular mechanism through which prefrontal circuits reorganize to support long-term memory. Although the brain networks supporting memory were known to change over time, how particular neurons and pathways acquire a role in remote recall remained poorly understood. I discovered that neurons activated during learning direct the recruitment of additional prefrontal neurons into the network supporting later recall. These recruited neurons project to temporal association cortex, and this pathway becomes necessary for recall as memories mature. I further showed that branches of the same prefrontal population reaching different brain regions exert opposing effects on defensive behavior. Together, these findings connect local neuronal recruitment to the changing behavioral roles of long-range pathways.
- 2014 – 2020 **University of Minnesota, Minneapolis MN**
Ph.D., Neuroscience, minor Neuroengineering
Advisor: Esther Krook-Magnuson, Ph.D.
Characterizing hippocampal modulation of memory and pathology. My graduate work investigated how hippocampal circuits shape memory, affect, and epilepsy pathology. I showed that epileptogenic insults to the ventral hippocampal cause distinct cognitive and affective deficits relative to dorsally-targeted insults, clarifying subregion-specific contributions to disease. I also found that cerebellar stimulation, though clinically effective for seizures, disrupts hippocampal place cell stability and impairs memory. Together, these studies reveal how hippocampal vulnerability and long-range modulation influence memory circuits and disease mechanisms.
- 2013 – 2014 **National Institute of Mental Health, Bethesda MD**
Postbaccalaureate trainee
Advisor: Carolyn Beebe Smith, Ph.D.
- 2008 – 2012 **The Colorado College, Colorado Springs CO**
B.A., Neuroscience, *cum laude*
Advisor: Lori Driscoll, Ph.D.

GRANTS & FELLOWSHIPS

Extramural Sources

- 2023 – 2026 NIMH F32 MH133387
“Reorganization of cortical memory ensembles across time”
- 2018 – 2020 NINDS F31 NS103320
“Cognitive outcomes of optogenetic seizure intervention in epileptic mice”

Intramural Sources

- 2022 – 2023 NIH T32 Training in Neurotechnology Translation Program, UCLA
“Temporal evolution of cortical memory ensembles”

2015 – 2017 NSF IGERT Neuroengineering
 “The HippoBellum: Understanding Cerebellar Control Over Hippocampal Functioning”

AWARDS & HONORS

2026 Brain Research Institute Annual Scheibel Postdoctoral Lecture Award, UCLA
2025 Schweizer Professional Development Award, UCLA
2022 Brain Research Travel Award, UCLA
2019 American Legion Brain Sciences Award
2018 President’s Student Leadership and Service Award, University of Minnesota
2017 Biomedical Graduate Education and Research Travel Award, University of Minnesota
2015 Morris-Smithberg Memorial Prize, University of Minnesota

MANUSCRIPTS & PUBLICATIONS

Pre-print / Under Review

Zeidler, Z., Yu, A., Seong, M., Hundley, R., DeNardo, L. Directed cell-type recruitment during consolidation of a remote memory (*bioRxiv; revised and resubmitted, Nature Neuroscience*)
<https://www.biorxiv.org/content/10.1101/2025.06.13.659562v1>

Goodpaster, C.M., **Zeidler, Z.**, Gongwer, M.W., Bland, C., Shari, M., Klune, C.B., Jones, N.S., Ramirez, M., Alturki, M.-B., Hundley, R., Utter, J., & DeNardo, L.A. Early adversity promotes adolescent avoidance behavior by enhancing prefrontal-amygdala communication through CRH⁺ glutamatergic neurons (*in submission*).

Peer-Reviewed Publications

de Sousa, A., **Zeidler, Z.**, Almeida-Filho, D., Shen, Y., Luchetti, A., Simanian, S., Madini, M., DeNardo, L., Silva, A. The prefrontal cortex controls memory organization in the hippocampus. *Nature Neuroscience*. 2026. doi: 10.1038/s41593-026-02231-1

Zeidler, Z., Fernandez-Gomez, M., Gupta, T., Shair, M., Wilke, S., DeNardo, L. Prefrontal dopamine activity is critical for rapid threat avoidance learning. *Current Biology*. 2025. doi:10.1016/j.cub.2025.07.035

Zeidler, Z. & DeNardo, L. 2024. Time-dependent transformations of prefrontal memory ensembles. In S. Ramirez & J. Graff (Eds), *The Engram handbook* (Nature-Springer). 2024.

Gabriel CJ, **Zeidler Z.**, Jin B, Guo C, Goodpaster CM, Kashay AQ, Wu A, Delaney M, Cheung J, DiFazio LE, Sharpe MJ, Aharoni D, Wilke SA, DeNardo LA. BehaviorDEPOT is a simple, flexible tool for automated behavioral detection based on markerless pose tracking. *eLife* 2022. doi:11:e74314. doi:10.7554/eLife.74314

Zeidler, Z., Hoffmann, K., Krook-Magnuson, E. HippoBellum: acute cerebellar modulation alters hippocampal dynamics and function. *Journal of Neuroscience*. 2020. DOI:10.1523/JNEUROSCI.0763-20.2020

Zeidler, Z., Brandt-Fontaine, M., Leintz, C., Krook-Magnuson, C., Netoff, T., KrookMagnuson, E. Targeting the mouse ventral hippocampus in the intrahippocampal kainic acid model of temporal lobe epilepsy. *eNeuro*. 2018. DOI: eneuro.0158–18.2018.

Qin, M., **Zeidler, Z.**, Moulton, K., Krych, L., Xia, Z., Smith, C.B. Endocannabinoid-mediated improvement on a test of aversive memory in a mouse model of fragile X syndrome. Behavioural brain research. 2015.
DOI:10.1016/j.bbr.2015.05.003

Qin, M., Huang, T., Krych, L., Xia, Z., Burlin, T., **Zeidler, Z.**, Zhao, T., Smith, C.B. R-baclofen reverses a social behavior deficit and elevated protein synthesis in a mouse model of fragile X syndrome. International Journal of Neuropsychopharmacology. 2015. DOI: 10.1093/ijnp/pyv034

Qin, M., Huang, T., Kader, M., Burlin, T., Xia, Z., **Zeidler, Z.**, Hukema, R.K., Smith, C.B. Cerebral Protein Synthesis in a Knockin Mouse Model of the Fragile X Premutation. ASN Neuro. 2014. DOI: 10.1177/1759091414551957

Non-Peer-Reviewed Publications

Zeidler, Z. & Krook-Magnuson, E. Back to the Dentate: Network Changes of Early- and Late Born Dentate Granule Cells. Epilepsy Currents. 2018.

Zeidler, Z. & Krook-Magnuson, E. One Site to Rule Them All: Toward a Master Regulator of Ictal Activity. Epilepsy Currents. 2016.

SELECTED TALKS

2026	Brain Research Institute Annual Scheibel Lecture, Los Angeles, CA
2026	Mount Sinai Postdoc Lecture Series, New York City, NY
2025	Society for Neuroscience Minisymposium, San Diego CA
2025	Pavlovian Society Annual Meeting, Sydney Australia (<i>declined due to federal funding freeze</i>)
2024	Gordon Research Seminar Optogenetic Approaches to Understanding Neural Circuits and Behavior, Lucca Italy
2024	UCLA Physiology Department Retreat, Los Angeles, CA
2024	Invited Speaker, Icahn School of Medicine at Mount Siani, New York City, NY
2024	Hatos Lecture Series, Los Angeles, CA
2018	University of Minnesota Graduate Program Annual Retreat, Minneapolis MN

TEACHING & LECTURES

2023 – 2025	Lecturer, 'Biology of Learning and Memory', UCLA Course director: Alcino Silva, Ph.D
2017 – 2019	Lecturer, 'Human Neuroanatomy', University of Minnesota Course director: Steven McLoon, Ph.D.
2017	Teaching Assistant, 'Human Neuroanatomy, University of Minnesota Course director: Steven McLoon, Ph.D.

ACADEMIC SERVICE & LEADERSHIP

2022 – 2023	Co-Founder, UCLA SYNCS External postdoc seminar series (Seminars by Young Neuroscience Community Scholars) that emphasizes both research excellence and community contribution through open-source tool development and/or non-academic community engagement
2020 – 2022	Chair, UCLA Postdoctoral Research Union, UAW Local 5810 Advocated for improved working conditions for over 1000 postdoctoral scholars including during contract negotiations with University of California leadership
2020 – 2022	Co-Founder, UCLA Physiology Trainee Seminar Series

2016 Co-President, Career Facilitation Committee, University of Minnesota
2015 – 2017 Elected Representative, University of Minnesota Graduate Student Board

Professional Memberships

2023 – present Pavlovian Society
2013 – present Society for Neuroscience

COMMUNITY OUTREACH

2023 Los Angeles County City of STEM Maker Faire, Los Angeles CA
2017 Science Museum of Minnesota, 'Social Science: Undead Rising', St. Paul MN
2017 State Fair, 'Brain Booth', St. Paul, MN
2016 Science Museum of Minnesota, 'Social Science: Eye Candy', St. Paul MN

ADVISING & MENTORING

2024 Abigail Yu, graduate student, UCLA
Rotation student, 'Effects of silencing prefrontal engrams on long-term memory'

2022 – 2025 Rio Hundley, undergraduate, UCLA
Directed research, 'Temporal changes in prefrontal encoding of fear memory'
**winner of Science Dean's Undergraduate Research Prize*

2022 – 2023 Meelan Shari, undergraduate, UCLA
Directed research, 'Quantification of retrograde targets from temporal association cortex'

2021 – 2023 Moonkyu (Patrick) Seong, undergraduate, UCLA
Directed research, 'Quantification of prelimbic projections to temporal association cortex'

2017 – 2018 Mikaela Brandt-Fontaine, undergraduate, University of Minnesota
Directed research, 'Inhibition of the vermal cerebellum to examine hippocampal connection via c-Fos expression'

2016 – 2018 Katerina Hoffman, undergraduate, University of Minnesota
Directed research, 'Hippocampal modulation in response to optogenetically stimulated cerebellar vermis'

2017 – 2018 Anmol Jaguste, undergraduate, University of Minnesota
Directed research, 'Semi-automated MATLAB-based seizure analysis'